Experiment 12

Implement echo client server using TCP/UDP sockets.

```
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec  7 2023, 22:03:25) [MSC v.1937 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\kishore\AppData\Local\Programs\Python\Python312\Ex 12(1) CN.py
Enter message: hello
Echo: hello
Enter message: exit
>>>
```

```
import socket
import threading
import time

def server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(("127.0.0.1", 5001))

    while True:
        data, addr = s.recvfrom(1024)
        if data.decode() == "exit":
            break
        s.sendto(data, addr)

    s.close()

threading.Thread(target=server, daemon=True).start()
time.sleep(1)

c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

while True:
    msg = input("Enter message: ")
    c.sendto(msg.encode(), ("127.0.0.1", 5001))
    if msg == "exit":
        break
    data, addr = c.recvfrom(1024)
    print("Echo:", data.decode())

c.close()
```

```
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec  7 2023, 22:03:25) [MSC v.1937 64 bit (AMD64)] on win32
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>>>
= RESTART: C:\Users\kishore\AppData\Local\Programs\Python\Python312\Ex 12(1) CN.py
Enter message: hello
Echo: hello
Enter message: exit
>>>
```

```
import socket
import threading

def server():
    s = socket.socket()
    s.bind(("127.0.0.1", 5000))
    s.listen(1)
    c, addr = s.accept()

    while True:
        msg = c.recv(1024).decode()
        if msg == "exit":
            break
        c.send(msg.encode())

    c.close()
    s.close()

threading.Thread(target=server, daemon=True).start()

c = socket.socket()
c.connect(("127.0.0.1", 5000))

while True:
    msg = input("Enter message: ")
    c.send(msg.encode())
    if msg == "exit":
        break
    print("Echo:", c.recv(1024).decode())

c.close()
```

Result: Given experiment is successfully completed.